SOLUTION

SECTION–I

1. Ans. (B)

Sol. $f(x) = 1 + x^n$ {put x = 3}

$1 + 3^n = -26$ (Rejected)

or

$f(x) = 1 - x^n$ {put x = 3}

$1 - 3^n = -26$

$3^n = 27$

$n = 3$

So $f(x) = 1 - x^3$

$f(4) = 1 - 4^3 = -63$

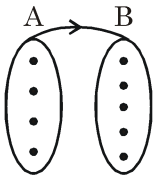
2. Ans. (C)

Sol. $f(x) + (x - 6) = (x-1)(x-2)(x-3)(x-4)(x-5)$

$\Rightarrow f(6) = 120$

3. Ans. (C)

Sol.



* $x = 5 \cdot 5 \cdot 5 \cdot 5 = 625$
* $y = {}^5C_4 . 4! = 120$

$\therefore$ $x + y = 625 + 120 = 745$

4. Ans. (B)

Sol. $y = \dfrac{2|x|-1}{3|x|+2} \Rightarrow |x| = \dfrac{-2y-1}{3y-2} \geq 0$

$\Rightarrow \dfrac{2y+1}{3y-2} \leq 0$

$\Rightarrow y \in \left[-\dfrac{1}{2}, \dfrac{2}{3}\right)$

5. Ans. (B)

Sol. $x^2 - 4x > 0, \quad x > 0$

$x(x-4) > 0$

$\Rightarrow x > 4$

6. Ans. (C)

Sol. $\cot^{-1}\dfrac{n}{\pi} > \dfrac{\pi}{6}$

$\dfrac{n}{\pi} < \sqrt{3} \Rightarrow n < \left(\sqrt{3}\pi\right) \approx 5.44$

$n = 5$

7. Ans. (B)

Sol. * $\sin^{-1}\left(\dfrac{1}{2}\right) = \dfrac{\pi}{6}$

* $\cos^{-1}\left(\dfrac{\sqrt{3}+1}{2\sqrt{2}}\right) = \dfrac{\pi}{12}$

* $\tan^{-1}\left(2-\sqrt{3}\right) = \dfrac{\pi}{12}$

$\therefore \sin^{-1}\dfrac{1}{2} + \cos^{-1}\dfrac{\sqrt{3}+1}{2\sqrt{2}} + \tan^{-1}\left(2-\sqrt{3}\right) = \dfrac{\pi}{3}$

8. Ans. (A,C,D)

Sol. (A) $f(x) = \dfrac{x^3}{(2^{|x|} + \sec x)}$ ...(1)

$f(-x) = \dfrac{-x^3}{(2^{|x|} + \sec x)}$ ...(2)

$f(x) + f(-x) = 0$ (odd function)

(B) Even function

(C) $f(x) = \dfrac{(e^x - 1)}{(e^x + 1)}(g(x) + g(-x))$ ...(1)

$f(x) = \dfrac{-(e^x - 1)}{(e^x + 1)}(g(x) + g(-x))$ ...(2)

(1) + (2) $\Rightarrow f(x) + f(-x) = 0$

(D) odd function.

9. Ans. (A,C,D)

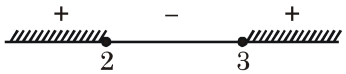
Sol. Let $|x| = t$

$\therefore \sqrt{t^2 - 5t + 6} = \sqrt{-t^2 + 2t + 8}$

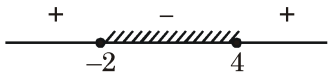
$\Rightarrow \sqrt{(t-3)(t-2)} = \sqrt{-(t-4)(t+2)}$

$\because (t-3)(t-2) \geq 0$ & $-(t-4)(t+2) \geq 0 \Rightarrow$

$(t-4)(t+2) \leq 0$



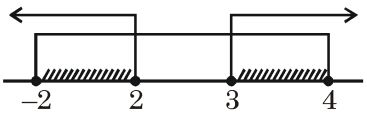
...(1)



...(2)

(1) $\cap$ (2)

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$-2 \le t \le 2 \cup 3 \le t \le 4$

$\Rightarrow |x| \le 2 \cup 3 \le |x| \le 4$

$\Rightarrow x \in [-2, 2] \cup [3, 4] \cup [-4, -3]$

10. Ans. (A,B,D)

Sol. $2\cot^{-1} 7 + \cos^{-1}\frac{3}{5} \Rightarrow 2\tan^{-1}\frac{1}{7} + \tan^{-1}\frac{4}{3}$

$\Rightarrow \tan^{-1}\left(\frac{7}{24}\right) + \tan^{-1}\left(\frac{4}{3}\right)$

$\{\because x > 0, y > 0, xy < 1\}$

$\Rightarrow \tan^{-1}\left[\frac{\frac{7}{24}+\frac{4}{3}}{1-\frac{7}{24}.\frac{4}{3}}\right]$

$\Rightarrow$

$\tan^{-1}\frac{117}{44} = \cot^{-1}\frac{44}{117} = \text{cosec}^{-1}\frac{125}{117} = \cos^{-1}\frac{44}{125}$

11. Ans. (A,D)

Sol. $\cos\left(\frac{\pi}{2} + \cos^{-1} x\right)$

$= -\sin\left(\cos^{-1} x\right) = -\sqrt{1-x^2} = -\frac{\sqrt{24}}{5}$

Paragraph for Question 12 to 14

12. Ans. (B)

Sol. $\frac{g(f(x))}{h(f(x))} = y = \frac{x^2-4x+5}{x^2-4x-12}$

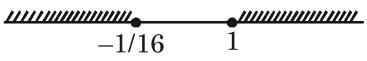
$\Rightarrow (y-1)x^2 + (4-4y)x - (12y+5) = 0$

$D \ge 0$

$\Rightarrow (4-4y)^2 + 4(y-1)(12y+5) \ge 0$

$\Rightarrow 16y^2 - 15y - 1 \ge 0$

$(y-1)(16y+1) \ge 0$



But at $y = 1$

$1 = \frac{x^2-4x+5}{x^2-4x-12}$

$\Rightarrow -12 = 5$ Not possible

$\therefore$ Range of $y: \left(-\infty, -\frac{1}{16}\right] \cup (1, \infty)$

$\therefore$ Integral values : $\{0, 1\}$

13. Ans. (C)

Sol. $g(f(x)) = x^2 - 4x + 5.$

Largest interval for invertible : $\left[-\frac{B}{2A}, \infty\right)$

$\therefore \quad a = \frac{-(-4)}{2.1} = 2$

Least value of $h(f(x)) = x^2 - 4x - 12$

$b = -\frac{(16+48)}{4} = -16$

$\therefore \sin^{-1}\left(\frac{8}{-16}\right) + \cos^{-1}\left(\frac{-8}{-16}\right) + \tan^{-1}\left(\frac{-32}{2\times(-16)}\right)$

$= \sin^{-1}\left(-\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right) + \tan^{-1}(1)$

$= -\frac{\pi}{6} + \frac{\pi}{3} + \frac{\pi}{4} = \frac{\pi}{4} + \frac{\pi}{6} = \frac{5\pi}{12}$

14. Ans. (C)

Sol. $\lim_{x\to 7}\lim_{n\to\infty}\frac{(x-7)^{2n}+1}{x^2-4x-12}$

$\lim_{x\to 7}\lim_{n\to\infty}\frac{((x-7)^2)^n+1}{x^2-4x-12} = \lim_{x\to 7}\frac{1}{(x-6)(x+2)}$

$= \frac{1}{1.9} = \frac{1}{9}$

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SECTION–III

1. Ans. 2

Sol. $f(x) = \text{sgn}[\ell n(\underbrace{(x-1)^2+1}_{1 \text{ to } \infty})]$

$= \text{sgn}[\ell n(1 \text{ to } \infty)]$

$= \text{sgn}[0 \text{ to } \infty]$

$\Rightarrow \text{sgn}(0) = 0$ & $\text{sgn}(+\text{ve}) = 1$

Number of integers in the range is 2.

2. Ans. 2

Sol. $f(x) = \dfrac{1}{(1+\cot x)} = \dfrac{\tan x}{(1+\tan x)}$;

$$f\left(\frac{\pi}{2}-x\right) = \frac{1}{(1+\tan x)} ;$$

$$f\left(\frac{\pi}{2}+x\right) = \frac{1}{1+\cot\left(\frac{\pi}{2}+x\right)} = \frac{1}{(1-\tan x)}$$

$$\text{and } f(\pi - x) = \frac{1}{(1-\cot x)}$$

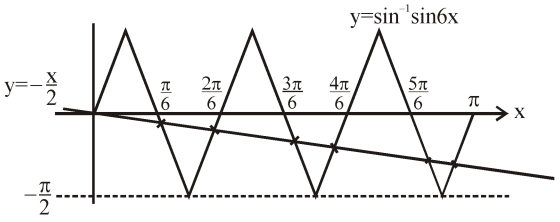
$$\Rightarrow \quad f(x) + f\left(\frac{\pi}{2}-x\right) + f\left(\frac{\pi}{2}+x\right) + f(\pi - x)$$

$$= \left[\frac{\tan x}{1+\tan x} + \frac{1}{1+\tan x}\right] + \left[\frac{1}{1-\tan x} + \frac{1}{1-\cot x}\right]$$

$= 1 + 1 = 2$

3. Ans. 7

Sol.



4. Ans. 2

Sol. $-1 \le x - 1 \le 1$ and $2 - x > 0$

$0 \le x \le 2$ and $x < 2$

integral values = {0, 1}

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